

Cellular Network

- Govt. regulatory agencies could not make spectrum allocations in proportion to the increasing demand for mobile services
- Restructuring of radio telephone system became imperative
 - To achieve high capacity
 - With limited radio spectrum
 - Covering very large areas

The cellular concept

- Replace a single high power transmitter (large cell) with many low power transmitter (small cells)
- Each cell provides coverage to only a small portion of service area
- Cellular concept was a major breakthrough in solving
 - Spectral congestion
 - User capacity

Cellular Architecture

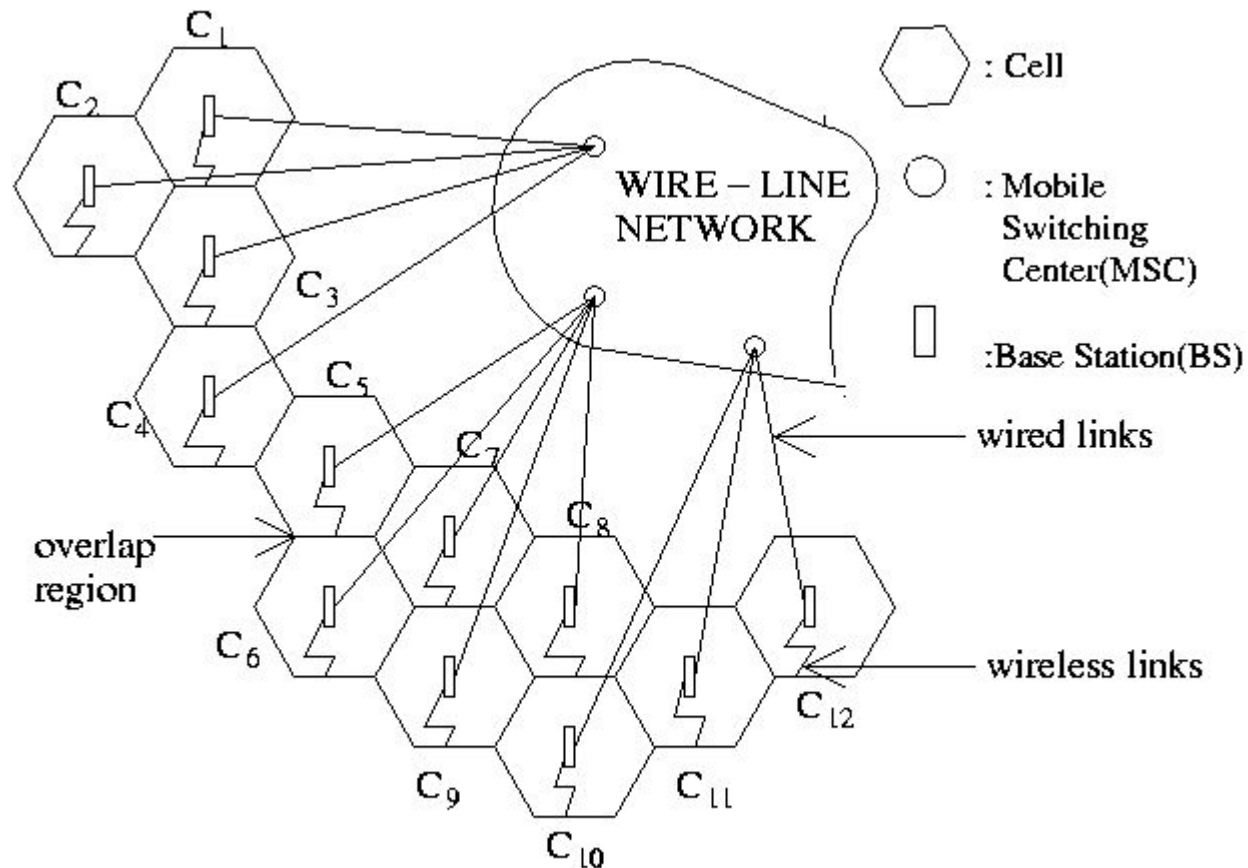


Fig. 1 : A Cellular Mobile Architechture

terminology

- Base Station (BS)
 - A fixed station in a mobile radio system
 - Used for radio communication with MS
 - Located at the centre or on the edge of a coverage region
 - Consists of
 - Radio channels
 - Transmitter
 - Receiver
 - antennas

terminology

- Mobile Switching Centre (MSC)
 - Co-ordinates routing of calls in a large service area
 - Connects BS and MSs to the PSTN
 - Also called mobile telephone switching office (MTSO)
- Transceiver
 - Device capable of simultaneously transmitting and receiving radio signals

terminology

- Mobile Station (MS)
 - Contains
 - Transceiver
 - Antenna
 - Control circuitry
 - Mounted in a vehicle
 - Used as a portable hand-held unit
- Forward Voice Channel (FVC)
 - Channels used for voice transmission from BS to MS
- Reverse Voice Channel (RVC)
 - Channels used for voice transmission from MS to BS

terminology

- FCC/ RCC
 - Two channels responsible for initiating mobile calls
 - Involved in setting up a call
 - Often called set-up channels
 - FCC serve as beacons
 - Continuously broadcast all of the traffic requests for all MSs in the system
 - 5% of total channels available are control/ setup channels
- A typical MSC handles
 - 100,000 cellular subscribers
 - 5,000 simultaneous conversation at a time
 - billing and system maintenance functions

How a call is made ?

- When a cell phone is turned on
 - scans the group of FCC to determine the one with the strongest signal
 - MS monitors the FCC until the signal drops below usable level
- Call initiated by MS
 - A call initiation request is sent on RCC
 - The servicing BS receives this data
 - Received data is sent to the MSC
 - The MSC validates the request
 - Makes connection to the called party through PSTN
 - Instructs the BS and MS to move to an unused FVC and RVC

- Call initiated by land phone
 - MSC dispatches the request to all BSs
 - Request message is broadcast as a paging message over all of the FCC
 - The MS for which MIN is broadcast responds by identifying itself over RCC
 - Servicing BS relays the acknowledgement to MSC
 - MSC instructs the BS to move the call to an unused voice channel
 - BS signals the MS to change frequencies to an unused FVC/ RVC pair
 - An alert message is also sent by FCC to instruct MS to ring

- Role of MSC during Call in progress
 - apply special control signal to control MS by BS and MSC
 - adjusts the transmitted power of MS
 - maintain call quality by changing the channel of MS

Cell Shape

- Actual radio coverage of a cell is known as footprint determined from
 - field measurements
 - propagation prediction models
- Real footprint is amorphous in nature
- Regular cell shape is essential for
 - systematic system design
 - future growth

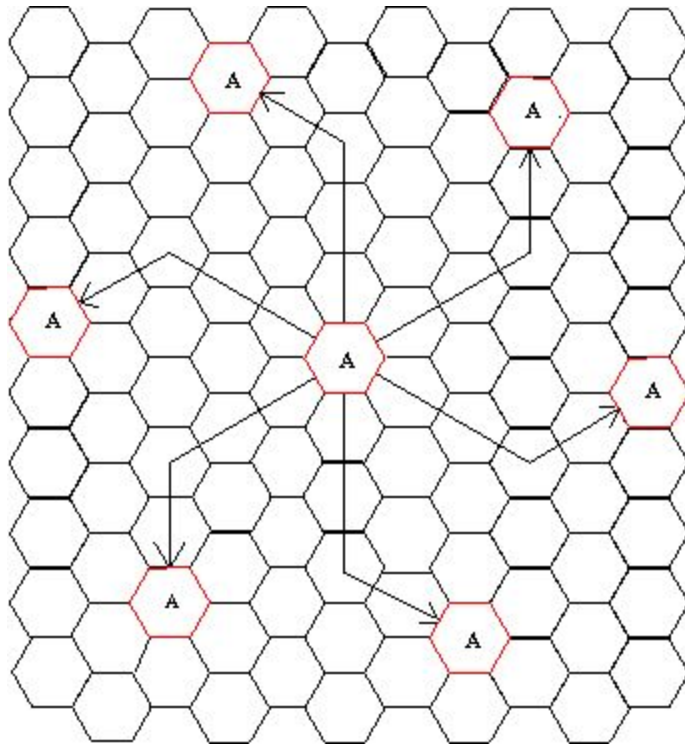
Cell Shape

- Regular shapes
 - rectangle
 - circle
 - hexagon
- For a given distance between the centre of a polygon and its farthest perimeter points
 - hexagon has the largest area of the three
 - hexagon closely approximates a circular radiation pattern matches with
 - omni-directional BS antenna
 - free-space propagation

Frequency reuse

- Place BSs systematically
- Distribute channel groups throughout the coverage zone
- Channels are reused as many times as necessary
 - Interference between co-channel cells should be below acceptable level

Co-channel cells



- Cells labeled with 'A' use the same group of channels
- No. of cells per cluster/ compact pattern N
$$N = i^2 + ij + j^2$$
$$i=3, j=2; N=19$$
- Each cell has 6 equidistant neighbour

Fig. 2 : Calculation of co-channel cells

Nearest co-channel neighbours

- Find the nearest co-channel cells
 - move i cells along any chain of hexagons
 - turn 60° counter-clockwise
 - move j cells
- Co-channel interference
 - Interference between signals of a cell and its co-channel cells is called co-channel interference

Cell Design

- Let a cellular system has
 - total of S duplex channels
 - each cell is allocated a group of K channels
 - N cells are in a cluster

$$S = KN$$

- A cluster is replicated M times
- Total no. of duplex channels

$$C \text{ (capacity)} = MKN = MS$$

Cell Design

- Capacity is directly proportional to the no. of times a cluster is replicated
- A large cluster size indicates
 - Ratio between the cell radius and the distance between co-channel cells D/R is large
- A small cluster size indicates
 - Co-channel cells are located much closer
 - Co-channel interference is significant

Cell Design

- Design criteria (over a given coverage)
 - maximise capacity
 - smallest possible value of N
 - no interference
 - biggest possible value of N
 - tradeoff between capacity and interference/QOS

Co-channel interference and system capacity

co-channel interference $Q = D/R$ (see Fig)

Q is too large

- Improves the transmission quality

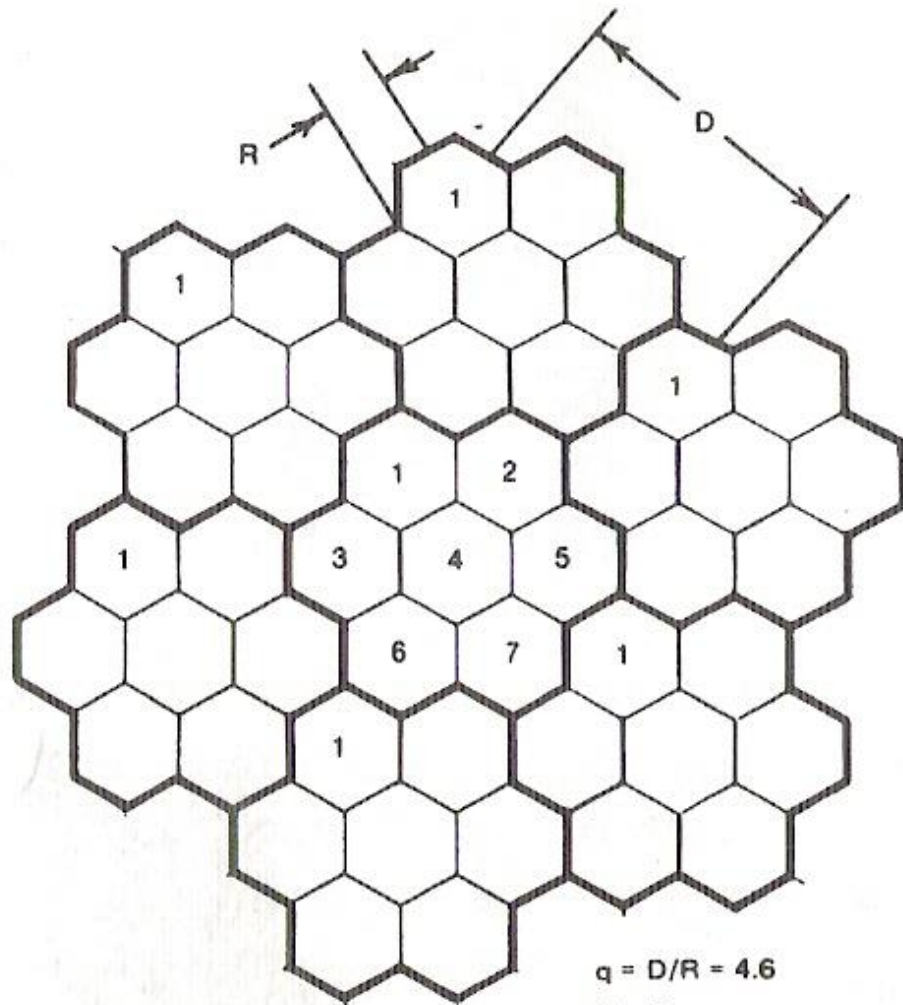
Q is too small

- Capacity increases

Trade-off must be made between quality and capacity

Reduce co-channel interference

- Separate co-channel cells by a minimum distance
- Provide sufficient isolation due to propagation



$$q = D/R = 4.6$$

$$K = 7$$

Hand Off strategies

- The process of transferring a mobile station from one channel or base station to another
 - Identify a new BS
 - Voice and control channels be allocated to channels associated with new BS
- Hand off must be performed
 - Successfully
 - As infrequently as possible
 - Imperceptible to the users

- In order to meet these requirements
 - System designer must specify an optimal signal level to initiate a hand off
 - Specify a particular signal level as minimum usable signal for acceptable voice quality at BS receiver
 - A slightly stronger signal level is used as threshold
 - This margin is given by

$$\text{delta} = P_{\text{rhandoff}} - P_{\text{rminimum usable}}$$

- Delta is too large
 - Unnecessary handoffs which burden the MSC may occur
 - Delta is too small
insufficient time to complete a handoff before
a call is lost due to weak signal

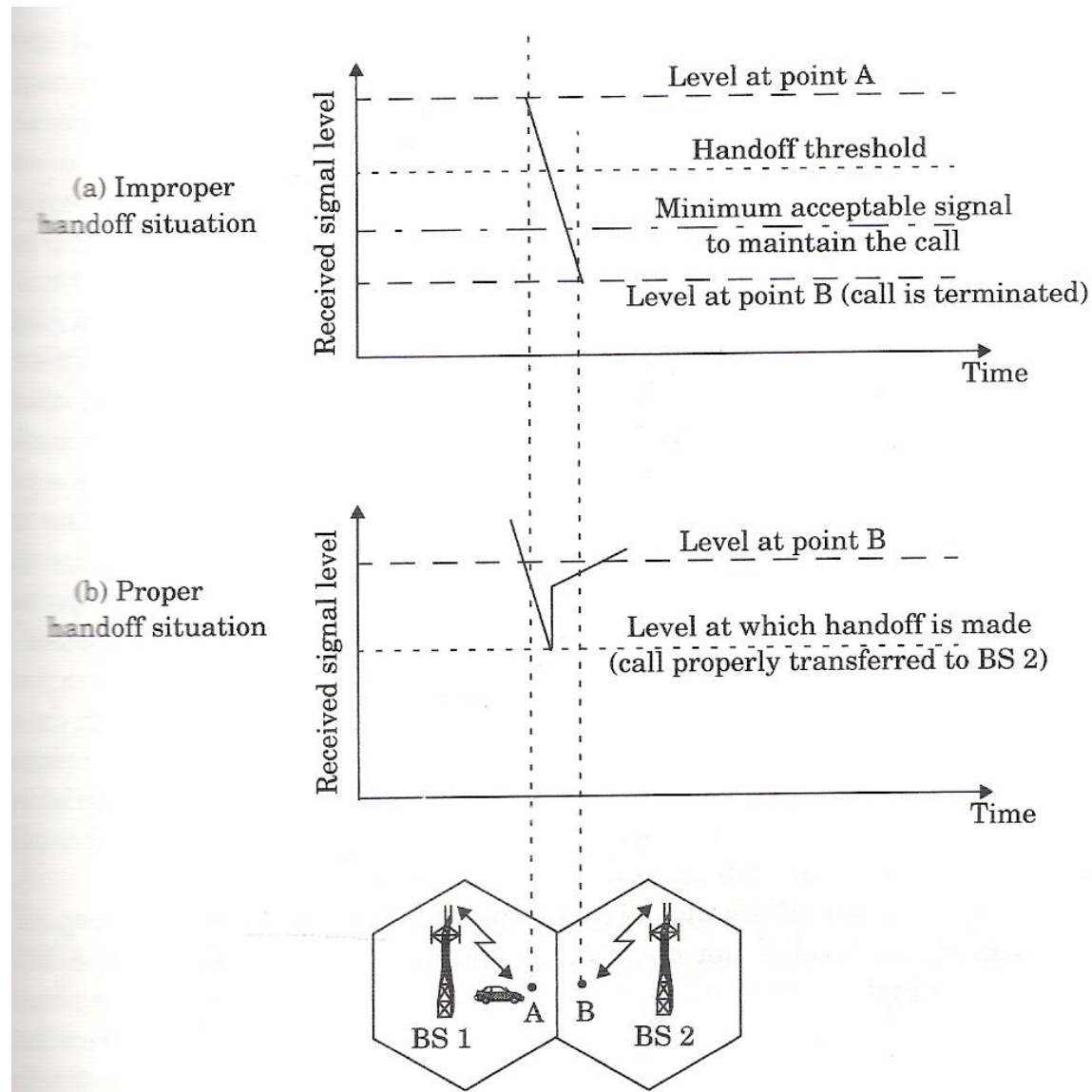


Fig. 3 : Illustration of a handoff scenario at cell boundary

- Call may be dropped for the following reasons
- Excessive delays may occur during high traffic condition for following reasons
 - computational loading at MSC
 - no channels are available on any of the nearby BSs
 - forces MSC to wait until a channel in a nearby cell becomes free

- Ensure the following before handoff
 - The drop in the measured signal level is not due to momentary fading
 - MS is actually moving away from the serving BS
- In order to ensure this
 - BS monitors the signal level for a certain period of time before a handoff is initiated
 - Duration of time depends on speed of MS
 - If the slope of the short-term average received signal level is steep, hand off should be made quickly

- 1st generation hand off
 - BS measures signal strength (ss)
 - MSC supervises
 - Each BS constantly monitors ss of RVCs
 - Determine relative location of each MS
- 2nd generation hand off
 - Hand off decisions are MS assisted (MAHO)
 - MS measures the received power from neighboring BSs
 - Continually reports the measurements to the serving BS

- A hand off is initiated
 - Power received from the BS of a neighbouring cell
> power received by the current BS by a certain
level or for a certain period of time
- Merits of 2nd gen hand off
 - Hand off is made by each MS
 - MAHO hand over a call much faster
 - MSC no longer constantly monitors ss

Roaming

- Roaming is a mechanism by means of which intersystem hand off takes place
- When roaming ?
 - Ss received by MS becomes weak
 - MSC can't find another cell within its system to transfer the control of MS
- What are the issues of roaming ?
 - MS moves out of its home system
 - Local call becomes long distance call
 - Compatibility between the two MSCs must be determined before implementing an intersystem hand off

Prioritizing hand off

User's view

- A call abruptly terminated while in the middle of a conversation is more annoying than being blocked occasionally on a new call attempt
- Hand off prioritization over new call attempt is desirable

Solution ?

- Guard channel concept
 - A fraction of channels is reserved for hand off
- Queuing of hand off requests

Hand off – some of the practical constraints

Problem 1

- Accommodating a wide range of mobile velocities during design

Solution ?

- Use different antenna heights
- Use different power levels
- Provide 'large' and 'small' cells co-located
- Known as umbrella cell approach (see Fig)
- Large/small area coverage to high/low speed users

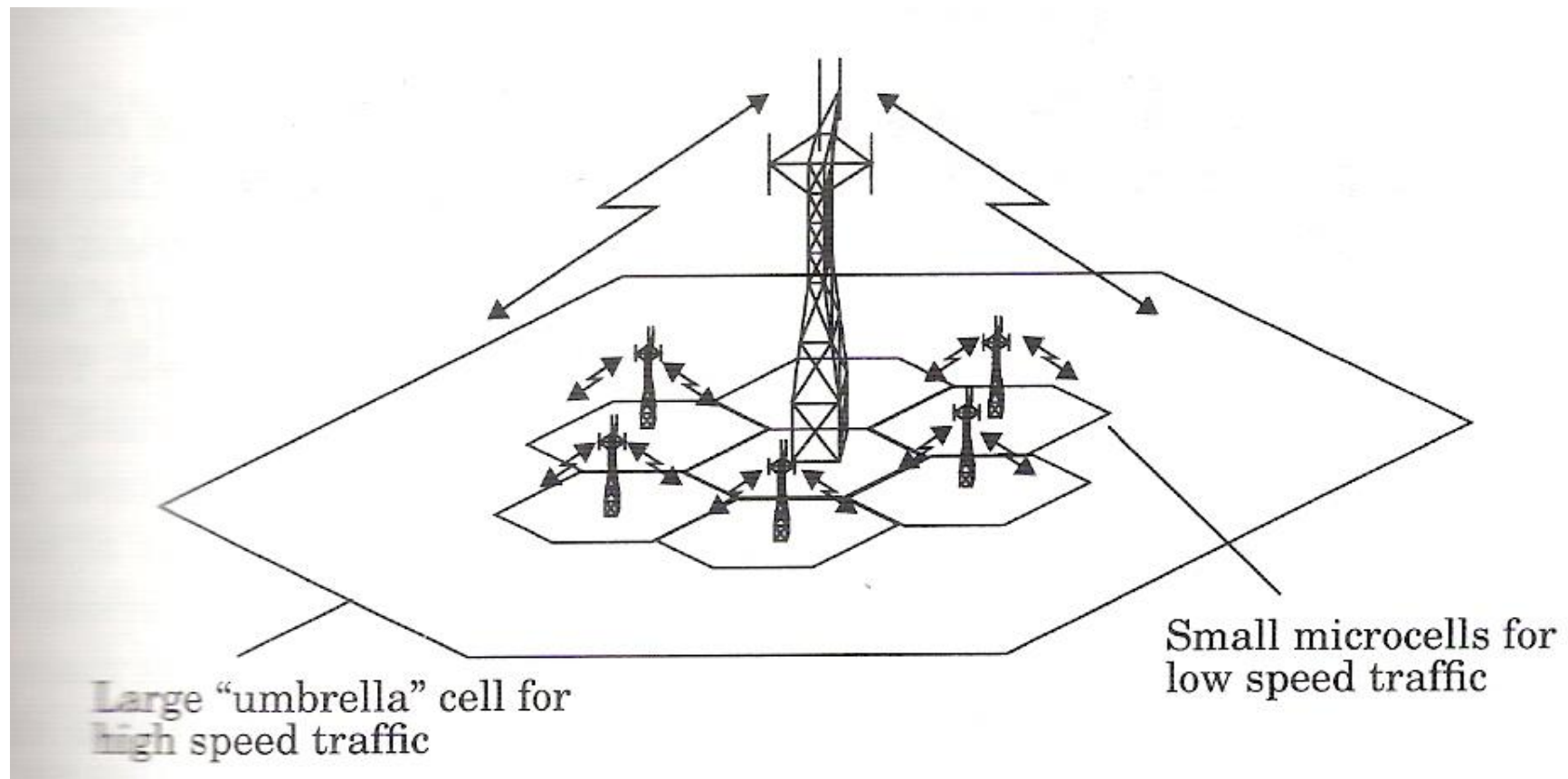


Fig. 4 : The umbrella cell approach

Problem 2

- Cell dragging
 - Hand off is not made even when essential
 - results from pedestrian users providing very strong signal to the BS
 - occurs in an urban environment
 - LOS radio path exists between MS and BS

GSM

- GSM (Global System for Mobile) is a 2nd generation cellular system standard
- World's first cellular system to specify digital modulation and network level architectures and services
- Before GSM, European countries used different cellular stds throughout the continent
 - Not possible for a customer to use a single subscriber unit throughout Europe

GSM contd.

- GSM services and features
 - Classified as either teleservices or dataservices
- Tele services
 - Calling
 - Facsimile
- Data services
 - Data rates from 300 bps to 9.6 kbps
 - SMS while simultaneously carrying normal voice traffic

GSM contd.

Remarkable features from the user's point of view

- SIM is a memory device that stores following information
 - subscriber's identification module
 - Networks
 - Country
 - Privacy key
 - Other user specific information

GSM contd.

- Without SIM installed all GSM mobiles are identical and nonoperational
- Subscriber may plug their SIM into any suitable terminal

GSM contd.

- GSM system architecture consists of three major interconnected subsystem
 - Base Station Subsystem (BSS)
 - Network and Switching Subsystem (NSS)
 - Operating Support Subsystem (OSS)

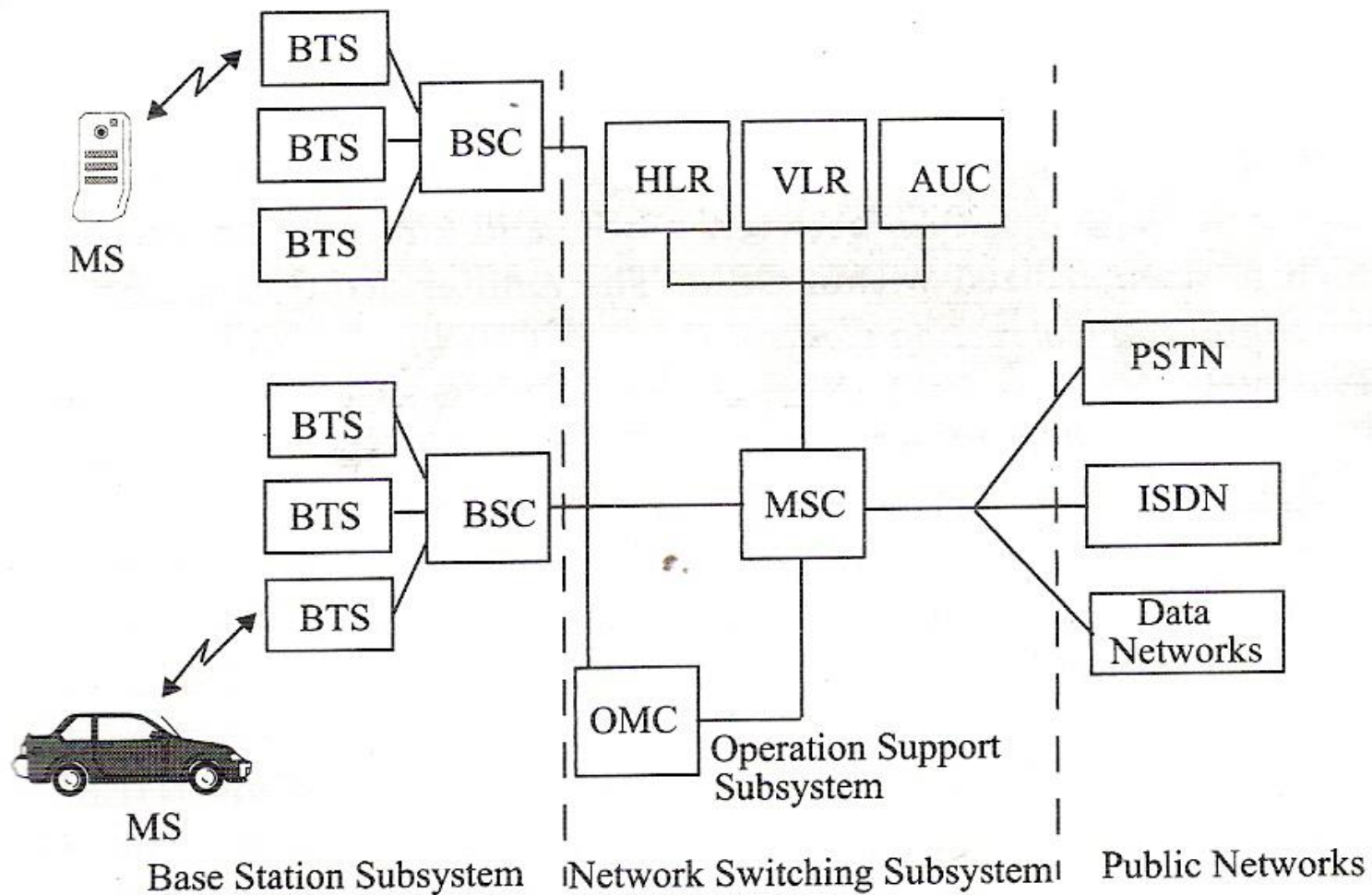


Fig. 5 GSM system architecture

GSM contd.

- BSS provides
 - Radio transmission paths between the MS and MSCs
 - Manages the radio interface between the MSs and all other subsystems of GSM
 - BSCs connect the MS to the NSS via MSCs
- NSS manages
 - Switching functions of the system
 - Allows the MSCs to communicate with other networks (PSTN, ISDN)

GSM contd.

- OSS supports
 - Operation and maintenance of GSM
 - Allows system engineers to monitor, diagnose and troubleshoot all aspects of the GSM
 - Solely for the staff of the GSM operating company
- Mobile handoffs between two BTSs under the control of the same BSC are handled by the BSC, and not the MSC

GPRS

GPRS (General Packet Radio Service)

- Multicast packet switched technology
- Enhanced 2nd generation cellular system with faster data service
- Particularly suited for sending and receiving small bursts of data such as email and web browsing

GPRS contd.

- Runs at speed up to 115 kbps compared with 9.6 kbps
 - Promises to support data transmission typically at 20 to 30 kbps (max 50 kbps)
 - Theoretically max up to 171.2 kbps
- Pay only for the amount of information you download rather than duration of the connection